PREP 4, LESSONS 7–8

Combining resistors and I–V graphs

Resistors in series and parallel (required practical); I–V characteristics

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Describe how to calculate the total resistance of two resistors in series. DN 118 [1]

Add the two resistances: $R_{total} = R_1 + R_2$

2 Describe the potential difference across components in parallel. DN 129 [1]

It is the same across each component.

3 Describe how the length of a wire affects its resistance. DN 112 [1]

The resistance increases (it is directly proportional to length).

Key ideas LESSONS 7–8

Show your working

4 Two 10 Ω resistors are connected in parallel.

a Their total resistance is [less than 10 Ω] | 10 Ω | 20 Ω [1]

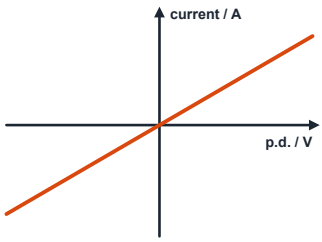
b Explain why adding a resistor in parallel decreases the total resistance. [2]

Each new branch gives the charge another path ✓

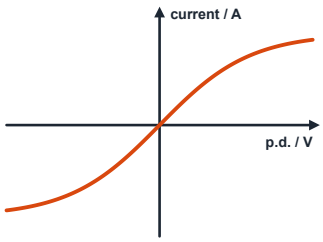
so the total current increases for the same p.d. ✓

5 These are the I–V characteristics of three components.

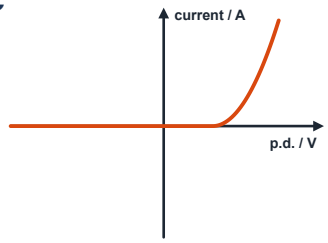
A



B



C



a Which graph shows a filament lamp? A | [B] | C [1]

b Which component lets current flow in one direction only? resistor | filament lamp | [diode] [1]

c Explain why the resistance of a filament lamp increases as the current increases. [2]

A greater current makes the filament hotter ✓

and its resistance increases with temperature ✓